

30/3/26

CO8 - AT8

S. Naveen
192425176Scenario Based

$$1. \text{ Image} = \begin{pmatrix} 45 & 48 & 46 \\ 47 & 220 & 49 \\ 46 & 48 & 47 \end{pmatrix} \text{ Threshold} = 500$$

$$\text{Mask} = \begin{pmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{pmatrix}$$

Step-1 Multiply corresponding values

$$\Rightarrow -45 - 48 - 46 - 47 + 1760 - 49 \\ - 46 - 48 - 47 \\ = 1384$$

$$|R| > T$$

$$|1384| > 500$$

2) Line detection:- It is a feature extraction technique to identify straight structure present in an image.

$$R = W \times f(x, y)$$

$$\text{Horizontal line Mask } H = \begin{pmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{pmatrix}$$

$$\text{Vertical line Mask } V = \begin{pmatrix} -1 & 2 & -1 \\ -1 & 2 & -1 \\ -1 & 2 & -1 \end{pmatrix}$$

45° diagonal line Mask $M = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$

-45° diagonal Mask $M = \begin{bmatrix} -1 & -1 & 2 \\ -1 & 2 & -1 \\ 2 & -1 & -1 \end{bmatrix}$

$$R = \sum_{i=-1}^1 \sum_{j=-1}^1 w(i,j) \times f(x+i, y+j).$$

$$R = 40 \quad T = 25$$

Horizontal line — Centre row has positive values

Vertical line — Centre column has positive values.

$$* \text{ Image} = \begin{bmatrix} 20 & 20 & 20 \\ 100 & 100 & 100 \\ 20 & 20 & 20 \end{bmatrix} \quad T = 200$$

1. compute the response of each Mask.
2. which Mask produces highest response.
3. what is the oriented of detected line.

Horizontal —

$$H = -20 - 20 - 20 + 200 + 200 + 200 - 20 \\ - 20 - 20$$

$$= -600 - 120 = -480.$$

$$\text{Vertical} = -20 + 40 - 20 - 100 + 200 - 100 - 20 \\ + 40 - 20 = 0.$$

$$D_1 = 40 - 20 - 20 - 100 + 200 - 100 - 20 - 20 + 40 = 0.$$

$$D_2 = -20 - 20 + 40 - 100 + 200 - 100 + 40 - 20 - 20 = 0.$$

Here, The management of highway offices detects that the highway has horizontal line.

Canny edge

1. Gaussian filtering

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

2. Gradient calculation.

$$G = \sqrt{G_x^2 + G_y^2}$$

3. Non-Maximum Suppression.

4. Double Thresholding.

5. Edge Tracking.

~~The input image is~~

Gradient calculation.

$$= \sqrt{(30)^2 + (40)^2}$$

$$= 60$$

$$P_1 - 42 \Rightarrow 55 > P_1 > 30$$

$$P_1 = 42 \rightarrow \text{weak edge.}$$

$$P_2 - 60 \rightarrow P_2 > 55 \rightarrow \text{strong edge.}$$

$$P_3 - 35 \rightarrow P_3 > 30 \text{ \& } P_3 < 55 = \text{weak edge.}$$

6) The edge will appear in final image after system becoz the weak edges are connected to strong edge.

7) Canny algorithms helps in way that an non clear image with rain noises.

Road land boundary detection

edge pixel	co-ordinates	Gradient direction.
P_1	(10, 10)	44°
P_2	(11, 11)	46°
P_3	(12, 12)	45°
P_4	(18, 18)	92°
P_5	(19, 19)	91°

The vehicle links two edges

pixel if euclidian distance is

two pixel.

Step-1 distance b/w P_1 and P_2

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$D = \sqrt{2} = 1.44$$

Step-2 linking condition formula.

find direction b/w P_1 & P_2

$$D = |144 - 46| = 2$$

$$P_1 - P_2 \Rightarrow d = \sqrt{2} = 1.44 \text{ \& } \Delta O = 1.$$

$$1.44 \leq 2$$

$$1 \leq 5$$

P_1 & P_2 are linked to each other.

$$2) P_2 - P_3 \Rightarrow d = \sqrt{2} = 1.414$$

$$1.414 \leq 2$$

$$1 \leq 5$$

$$3) P_3 - P_4 = \sqrt{72} = 8.48$$

$$8.48 \neq 2$$

$$45 \neq 5$$

$$4) P_4 - P_5 \Rightarrow \sqrt{2} = 1.414 \Delta O = 1.$$

$$1.414 \leq 2$$

$P_4 - P_5$ are linked to each other.

Problem understanding 20%	Identification of solution. 20%	Solution process 25%	Decision making 20%	Clarity of response 15%	Total 100%
20	20	25	20	15	
20	20	24	20	14	98